

Dhiraj Salunkhe

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EDUCATION

University of Pennsylvania

Master of Science in Engineering (M.S.E.) in Systems Engineering

Aug 2025 – May 2027
Philadelphia, PA

Coursework: Machine Learning, Robotics, Convex Optimization, Statistics for Data Science

Savitribai Phule Pune University

Bachelor of Engineering in Information Technology (First Class with Distinction)

Aug 2019 – Apr 2022
Pune, India

Coursework: Computer Vision, Deep Learning, Data Structures & Algorithms, Linear Algebra

EXPERIENCE

Research Assistant

The Wharton School, University of Pennsylvania

Jan 2026 – Mar 2026
Philadelphia, PA

- Implemented and trained transformer-based deep learning models in PyTorch from scratch across 15 experimental configurations, building data ingestion and preprocessing pipelines with train/test scene splits that prevented data leakage.
- Stood up evaluation harnesses measuring mIoU-style metrics for model accuracy and semantic quality, applying attention mechanisms and experiment tracking via MLflow to benchmark architecture choices and improve model accuracy by 3%.
- Presented model results and trade-offs to senior researchers through structured reports and dashboards, translating perception research findings into actionable engineering decisions.

Software Engineer

3-i Infotech (Client: CRISIL - S&P Global)

Oct 2022 – Jul 2025
Mumbai, India

- Implemented transformer-based deep learning architectures in PyTorch and TensorFlow for multi-view spatial data perception and 3D pattern detection across CRISIL's ratings and global market data platforms at S&P Global, benchmarking architecture choices against modern references and reducing data reprocessing by 30% and batch time by 25%.
- Built data ingestion and preprocessing pipelines compatible with annotation formats and sensor-adjacent data schemas, implementing train/test scene splits to prevent data leakage and standing up evaluation harnesses measuring mAP and mIoU-style detection metrics on held-out evaluation sets.
- Applied attention mechanisms and multi-view data fusion techniques using PyTorch transformers, optimizing models for real-time inference with ONNX and TensorRT-adjacent export tooling to meet production latency requirements.
- Tracked experiments with MLflow and Weights & Biases, performing GPU profiling to benchmark model performance and calibrate data pipelines for reliable, longer-range detection in production environments at S&P Global's market intelligence products.
- Led 10 engineers across multiple CRISIL perception-adjacent ML projects, presenting model results and trade-offs to company leadership; received the YMD Award for delivering critical platform upgrades ahead of schedule.

Software Engineer

CMS IT Services

Aug 2021 – Sept 2022
Mumbai, India

- Built deep learning data pipelines and evaluation harnesses in Python and PyTorch, resolving 4 critical production issues and improving system reliability from 99.5% to 99.9% uptime for 50,000+ active users.

TECHNICAL SKILLS

Perception & 3D: Transformers, Attention Mechanisms, BEV (Bird's-Eye-View), Multi-View Fusion, 3D Detection

Deep Learning: PyTorch (from scratch), TensorFlow, JAX, CNNs, ViT, Object Detection, Semantic Segmentation

Inference & GPU: ONNX, TensorRT (exposure), CUDA, GPU Profiling, Distributed Training, Mixed Precision

Robotics & Sensors: ROS (exposure), Point Cloud (PCL, Open3D exposure), LiDAR concepts, Sensor Calibration

MLOps: MLflow, Weights & Biases, Experiment Tracking, mIoU, mAP, Evaluation Harnesses

Languages: Python, C++ , CUDA, SQL, Bash

Cloud & Tools: AWS, Docker, Kubernetes, Git, GitHub Actions, Linux

PROJECTS

Transformer-Based Multi-View Perception Model with BEV Evaluation Harness | PyTorch, ViT, ONNX, MLflow, W&B

- Implemented a transformer-based multi-view perception architecture in PyTorch from scratch, benchmarking against BEV-style reference architectures for 3D object detection and semantic occupancy prediction with mIoU and mAP evaluation on held-out scenes.
- Built data ingestion and preprocessing pipeline with annotation-compatible format, train/test scene splits preventing data leakage, and ONNX export for real-time inference optimization, tracking experiments with MLflow and Weights & Biases.

3D Spatial Detection Pipeline with LiDAR-Adjacent Point Cloud Processing | Python, PyTorch, Open3D, TensorRT, CUDA

- Developed a 3D detection and semantic segmentation pipeline using PyTorch transformers and attention mechanisms for multi-view spatial data fusion, applying point cloud processing concepts and sensor calibration validation to improve detection reliability.
- Optimized model for real-time inference using TensorRT-adjacent export and GPU profiling with CUDA, standing up evaluation harnesses measuring mAP on held-out evaluation sets and presenting trade-off analysis to senior engineers.